Distinct real Eigenvalues

We saw previously that the solution to a system of two homogeneous differential equations $\vec{x}' = A\vec{x}$ (when A is 2×2), would be given by $\vec{x} = c_1\vec{x_1} + c_2\vec{x_2}$, where the solutions are vectors $\{\vec{x_1}, \vec{x_2}\}$ defined in t as $\vec{x_1} = \vec{k_1}e^{\lambda_1 t}$ and $\vec{x_2} = \vec{k_2}e^{\lambda_2 t}$.

Given that we know the form of these solution vectors, we want to turn our attention toward finding these solution vectors for the homogeneous system $\vec{x}' = A\vec{x}$.

As it turns out, these solution vectors will be defined by the Eigenvalues and Eigenvectors of the matrix A.

Eigenvalues and Eigenvectors

If we plug a solution vector $\vec{x} = \vec{k}e^{\lambda t}$ into the matrix equation $\vec{x}' = A\vec{x}$, we get

$$\vec{x}' = A\vec{x}$$

$$\vec{k}\lambda e^{\lambda t} = A\vec{k}e^{\lambda t}$$

$$\vec{k}\lambda = A\vec{k}$$

$$A\vec{k} - \vec{k}\lambda = \vec{O}$$

$$(A - \lambda I)\vec{k} = \vec{O}$$

© Krista King Math

To understand this equation, let's consider the case where A is a 2×2 matrix. Then the matrix $A - \lambda I$ is some 2×2 matrix A, minus the product of the constant λ and the 2×2 identity matrix I_2.

$$A - \lambda I = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} a_{11} - \lambda & a_{12} \\ a_{21} & a_{22} - \lambda \end{bmatrix}$$

So we can rewrite the equation $(A - \lambda I)\vec{k} = \vec{O}$ as

$$\begin{bmatrix} a_{11} - \lambda & a_{12} \\ a_{21} & a_{22} - \lambda \end{bmatrix} \vec{k} = \vec{O}$$

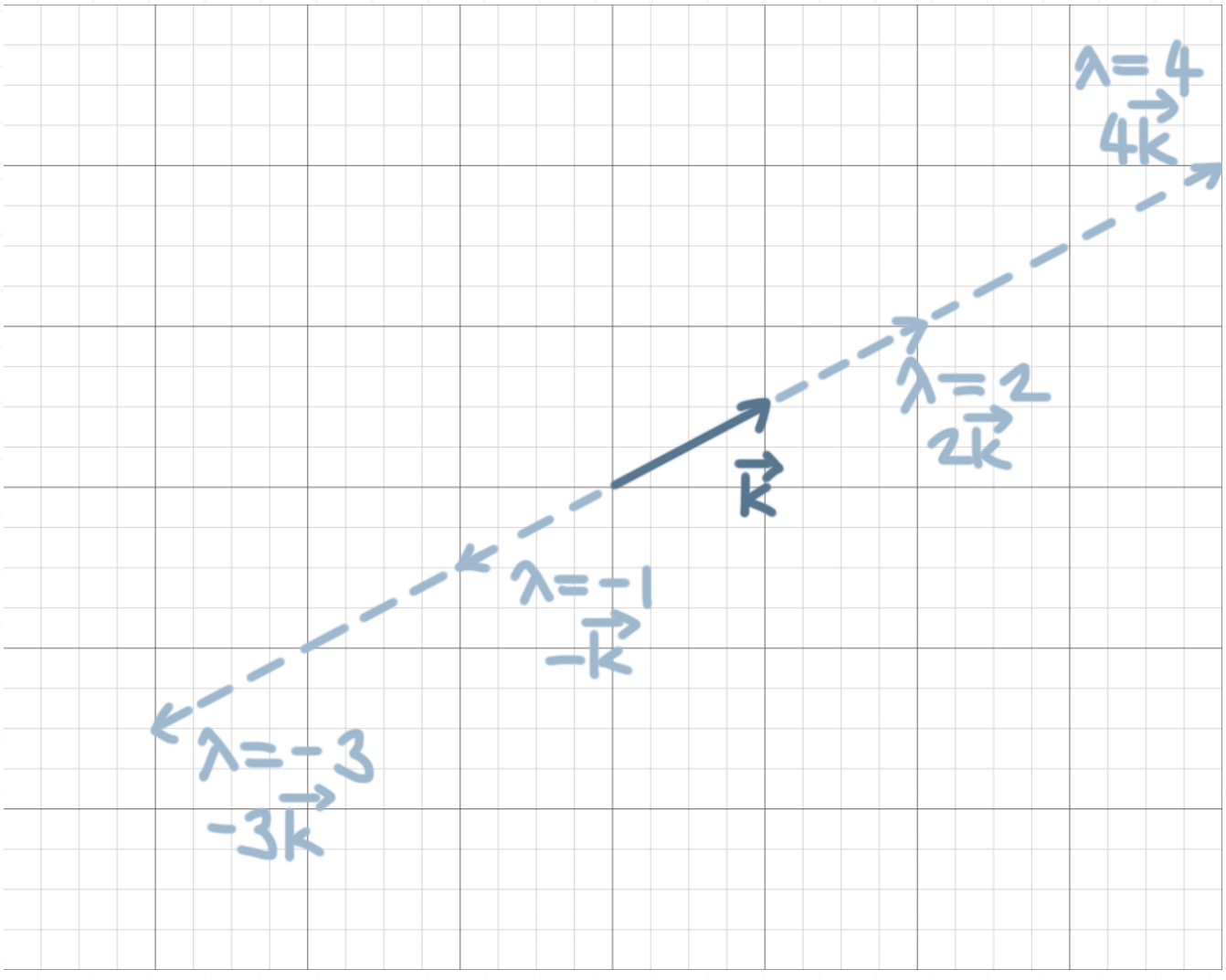
We're looking for solution vectors $\vec{k}$ that satisfy this equation. In other words, when we multiply the matrix $A - \lambda I$ by $\vec{k}$, we should get the zero vector $\vec{O}$.

Of course, the vector $\vec{k} = \vec{O}$ will always satisfy the equation, so we consider that to be the **trivial solution**. We're more interested in any non-zero vectors $\vec{k}$, any **nontrivial solutions**, that satisfy the equation.

What are these nontrivial solutions? Well, we saw before that, after plugging a solution vector $\vec{x} = \vec{k}e^{\lambda t}$ into the matrix equation $\vec{x}' = A\vec{x}$, we found $\vec{k}\lambda = A\vec{k}$. This equation tells us that, when we multiply the matrix A by the vector $\vec{k}$, the result is the same vector $\vec{k}$, just scaled by some constant λ. In other words, the result $\lambda\vec{k}$ will lie along the same "line" as the original $\vec{k}$, it'll just be some scaled version of $\vec{k}$.

© Krista King Math





The values of λ that make this equation true are the **Eigenvalues** of the matrix A, while the vectors $\vec{k}$ that satisfy it are the **Eigenvectors** of the matrix A. In other words, all of the non-trivial solutions we want to find are Eigenvalue/Eigenvector pairs for matrix A.

Therefore, our goal is to find any Eigenvalue/Eigenvector pairs for matrix A, and then use those $(\lambda, \vec{k})$ pairs to build solutions to the system $\vec{x}' = A\vec{x}$ in the form $\vec{x} = \vec{k}e^{\lambda t}$.

By definition, a homogeneous system will always either have exactly one solution (which will be the trivial solution), or it'll have infinitely many solutions. The trivial solution isn't interesting; we're looking for the "infinitely many solutions" case. We only get these non-trivial solutions when the matrix A is singular, which occurs when the determinant $|A - \lambda I|$ is 0,

© Krista King Math

$$|A - \lambda I| = 0$$

This determinant equation is called the **characteristic equation**, and solving it will give us the Eigenvalues of the matrix A, and then we can use the Eigenvalues to find the associated Eigenvectors.

Distinct real Eigenvalues

There are three cases we want to consider.

1. **Distinct real**: The matrix A has distinct real Eigenvalues (every Eigenvalue is a real number, and no Eigenvalues are equivalent).

2. **Equal real**: The matrix A has at least some equal real Eigenvalues (every Eigenvalue is a real number, but at least some of the Eigenvalues are equivalent).

3. **Complex conjugate**: The matrix A has complex Eigenvalues (the Eigenvalues are complex numbers).

In this lesson, we'll start with just the case of distinct real Eigenvalues. The Eigenvectors that correspond to distinct real Eigenvalues will always be linearly independent vectors, which means the solutions $\overrightarrow{x_1} = \overrightarrow{k_1}e^{\lambda_1 t}$ and $\overrightarrow{x_2} = \overrightarrow{k_2}e^{\lambda_2 t}$ we find for the system will always be linearly independent, and will therefore form a fundamental set of solutions.

Let's do an example where we use Eigenvalue/Eigenvector pairs for a set of distinct real Eigenvalues in order to solve a system of differential equations.

© Krista King Math



Example

Solve the system of differential equations.

$$x_1' = -6x_1 + 2x_2$$

$$x_2' = -3x_1 + x_2$$

The coefficient matrix is

$$A = \begin{bmatrix} -6 & 2 \\ -3 & 1 \end{bmatrix}$$

and the matrix $A - \lambda I$ is

$$A - \lambda I = \begin{bmatrix} -6 & 2 \\ -3 & 1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} -6 & 2 \\ -3 & 1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} -6 - \lambda & 2 \\ -3 & 1 - \lambda \end{bmatrix}$$

Its determinant is

$$\begin{vmatrix} -6 - \lambda & 2 \\ -3 & 1 - \lambda \end{vmatrix} = (-6 - \lambda)(1 - \lambda) - (2)(-3)$$

$$\begin{vmatrix} -6 - \lambda & 2 \\ -3 & 1 - \lambda \end{vmatrix} = -6 + 6\lambda - \lambda + \lambda^2 + 6$$

© Krista King Math

$$\begin{vmatrix} -6-\lambda & 2 \\ -3 & 1-\lambda \end{vmatrix} = \lambda^2 + 5\lambda$$

So the characteristic equation is

$$\lambda^2 + 5\lambda = 0$$

$$\lambda(\lambda + 5) = 0$$

$$\lambda = -5, 0$$

Then for these Eigenvalues, $\lambda_1 = -5$ and $\lambda_2 = 0$, we find

$$A - (-5)I = \begin{bmatrix} -6-(-5) & 2 \\ -3 & 1-(-5) \end{bmatrix}$$

$$A - (-5)I = \begin{bmatrix} -1 & 2 \\ -3 & 6 \end{bmatrix}$$

and

$$A - (0)I = \begin{bmatrix} -6-0 & 2 \\ -3 & 1-0 \end{bmatrix}$$

$$A - (0)I = \begin{bmatrix} -6 & 2 \\ -3 & 1 \end{bmatrix}$$

Put these two matrices into reduced row-echelon form.

$$\begin{bmatrix} -1 & 2 \\ -3 & 6 \end{bmatrix} \qquad \begin{bmatrix} -6 & 2 \\ -3 & 1 \end{bmatrix}$$

© Krista King Math



$$\begin{bmatrix} 1 & -2 \\ -3 & 6 \end{bmatrix} \qquad \begin{bmatrix} 1 & -\frac{1}{3} \\ -3 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix} \qquad \begin{bmatrix} 1 & -\frac{1}{3} \\ 0 & 0 \end{bmatrix}$$

If we turn these back into systems of equations (in both cases we only need to consider the equation that we get from the first row of each matrix), we get

$$k_1 - 2k_2 = 0 \qquad k_1 - \frac{1}{3}k_2 = 0$$

$$k_1 = 2k_2 \qquad k_1 = \frac{1}{3}k_2$$

From the first system, we'll choose $k_2 = 1$, which results in $k_1 = 2$. And from the second system, we'll choose $k_2 = 3$, which results in $k_1 = 1$.

$$\overrightarrow{k_1} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \qquad \overrightarrow{k_2} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

Then the solutions to the system are

$$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t} \qquad \overrightarrow{x_2} = \overrightarrow{k_2} e^{\lambda_2 t}$$

$$\overrightarrow{x_1} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} e^{-5t} \qquad \overrightarrow{x_2} = \begin{bmatrix} 1 \\ 3 \end{bmatrix} e^{0t}$$

$$\overrightarrow{x_1} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} e^{-5t} \qquad \overrightarrow{x_2} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

© Krista King Math

Therefore, the general solution to the homogeneous system must be

$$\vec{x} = c_1\vec{x_1} + c_2\vec{x_2}$$

$$\vec{x} = c_1\begin{bmatrix}2\\1\end{bmatrix}e^{-5t} + c_2\begin{bmatrix}1\\3\end{bmatrix}$$

Let's do one more example where we convert a second order equation into a system, and then solve an initial value problem with the system.

Example

Convert the differential equation into a system of equations, then use the system to solve the initial value problem, given $y(0) = 1$ and $y'(0) = -1$.

$$y'' - 5y' + 6y = 0$$

We can start by defining

$$x_1(t) = y(t)$$

$$x_2(t) = y'(t)$$

and then we'll solve the original differential equation for $y''(t)$.

$$y'' - 5y' + 6y = 0$$

$$y'' = 5y' - 6y$$

© Krista King Math



Then if we take the derivatives of $x_1(t)$ and $x_2(t)$, we get

$$x_1'(t) = y'(t) = x_2(t)$$

$$x_2'(t) = y''(t) = 5y' - 6y = 5x_2(t) - 6x_1(t)$$

Simplifying these equations gives us a system of equations that's equivalent to the original second order differential equation.

$$x_1' = x_2$$

$$x_2' = 5x_2 - 6x_1$$

We can represent this system in matrix form as

$$\vec{x}' = A\vec{x}$$

$$\vec{x}' = \begin{bmatrix} 0 & 1 \\ -6 & 5 \end{bmatrix} \vec{x}$$

and then find the matrix $A - \lambda I$,

$$A - \lambda I = \begin{bmatrix} 0 & 1 \\ -6 & 5 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 0 & 1 \\ -6 & 5 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} -\lambda & 1 \\ -6 & 5 - \lambda \end{bmatrix}$$

and its determinant $|A - \lambda I|$.

© Krista King Math

$$|A - \lambda I| = \begin{vmatrix} -\lambda & 1 \\ -6 & 5 - \lambda \end{vmatrix}$$

$$|A - \lambda I| = (-\lambda)(5 - \lambda) - (1)(-6)$$

$$|A - \lambda I| = -5\lambda + \lambda^2 + 6$$

$$|A - \lambda I| = \lambda^2 - 5\lambda + 6$$

Solve the characteristic equation for the Eigenvalues.

$$\lambda^2 - 5\lambda + 6 = 0$$

$$(\lambda - 2)(\lambda - 3) = 0$$

$$\lambda = 2, 3$$

For the Eigenvalue $\lambda_1 = 2$ we find

$$A - 2I = \begin{bmatrix} -2 & 1 \\ -6 & 3 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} -2 & 1 \\ -6 & 3 \end{bmatrix} \to \begin{bmatrix} 1 & -1/2 \\ -6 & 3 \end{bmatrix} \to \begin{bmatrix} 1 & -1/2 \\ 0 & 0 \end{bmatrix}$$

If we turn this first row into an equation, we get

$$k_1 - \frac{1}{2}k_2 = 0$$

$$k_1 = \frac{1}{2}k_2$$

© Krista King Math



We'll choose $k_2 = 2$, which results in $k_1 = 1$.

$$\vec{x_1} = \vec{k_1} e^{\lambda_1 t}$$

$$\vec{x_1} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{2t}$$

For the Eigenvalue $\lambda_1 = 3$ we find

$$A - 3I = \begin{bmatrix} -3 & 1 \\ -6 & 2 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} -3 & 1 \\ -6 & 2 \end{bmatrix} \to \begin{bmatrix} 1 & -1/3 \\ -6 & 2 \end{bmatrix} \to \begin{bmatrix} 1 & -1/3 \\ 0 & 0 \end{bmatrix}$$

If we turn this first row into an equation, we get

$$k_1 - \frac{1}{3} k_2 = 0$$

$$k_1 = \frac{1}{3} k_2$$

We'll choose $k_2 = 3$, which results in $k_1 = 1$.

$$\vec{x_2} = \vec{k_2} e^{\lambda_2 t}$$

$$\vec{x_2} = \begin{bmatrix} 1 \\ 3 \end{bmatrix} e^{3t}$$

So the general solution to the homogeneous system is

© Krista King Math

$$\vec{x} = c_1\vec{x_1} + c_2\vec{x_2}$$

$$\vec{x} = c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} e^{3t}$$

Remember that the initial conditions were given for the original differential equation as $y(0) = 1$ and $y'(0) = -1$. But when we started this problem we set

$$x_1(t) = y(t)$$

$$x_2(t) = y'(t)$$

If we substitute $t = 0$ into these equations, we get

$$x_1(0) = y(0) = 1$$

$$x_2(0) = y'(0) = -1$$

These new initial conditions are $x_1(0) = 1$ and $x_2(0) = -1$, which means the vector $\vec{x}(0) = (1, -1)$. Plugging this vector and $t = 0$ (since this is the value given in the initial conditions) into the general solution, we get

$$\begin{bmatrix} 1 \\ -1 \end{bmatrix} = \vec{x}(0) = c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{2(0)} + c_2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} e^{3(0)}$$

$$\begin{bmatrix} 1 \\ -1 \end{bmatrix} = \vec{x}(0) = c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} (1) + c_2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} (1)$$

$$\begin{bmatrix} 1 \\ -1 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

which gives us the system of equations

© Krista King Math

$$c_1 + c_2 = 1$$

$$2c_1 + 3c_2 = -1$$

Solving this system gives $c_1 = 4$ and $c_2 = -3$, so the solution to the initial value problem is therefore

$$\vec{x} = 4\begin{bmatrix}1\\2\end{bmatrix}e^{2t} - 3\begin{bmatrix}1\\3\end{bmatrix}e^{3t}$$

This is the solution to the system, but we'd like to find the solution to the original second order differential equation. If we just remember that we defined

$$\vec{x}(t) = \begin{bmatrix}x_1(t)\\x_2(t)\end{bmatrix} = \begin{bmatrix}y(t)\\y'(t)\end{bmatrix}$$

when we started this problem, then we can break down our solution into two equations,

$$x_1(t) = y(t) = 4e^{2t} - 3e^{3t}$$

$$x_2(t) = y'(t) = 8e^{2t} - 9e^{3t}$$

and the solution $y(t)$ to the original differential equation is therefore

$$y(t) = 4e^{2t} - 3e^{3t}$$

© Krista King Math